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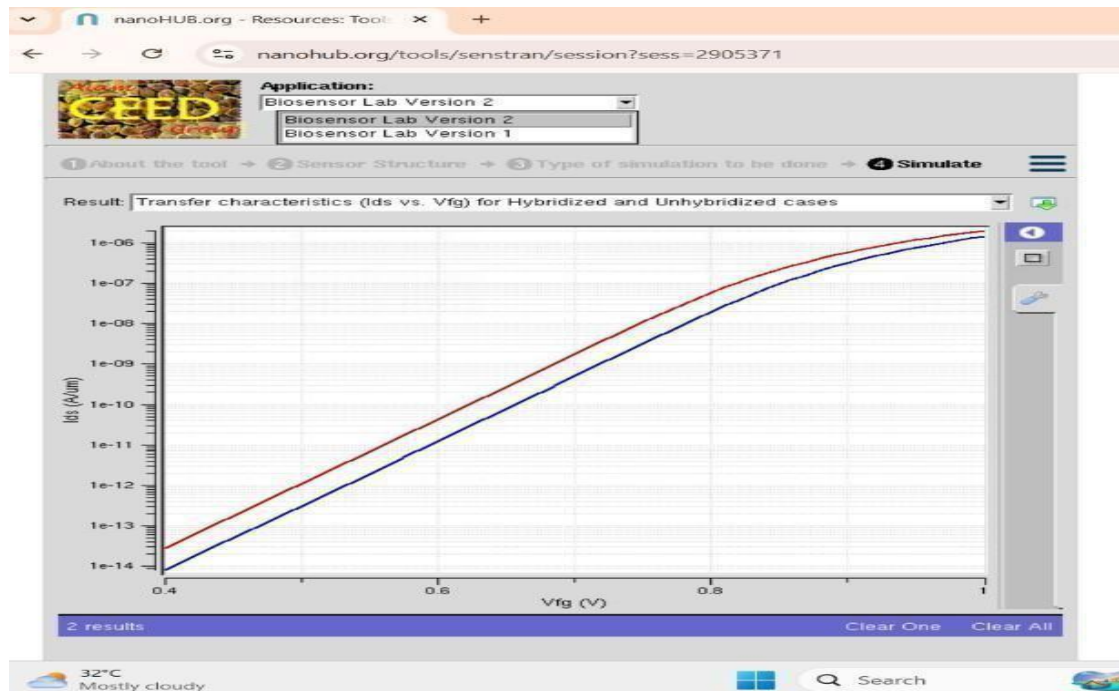
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Exp. 9. Biosensor and Bio-FET characteristics using NanoHub

Aim:

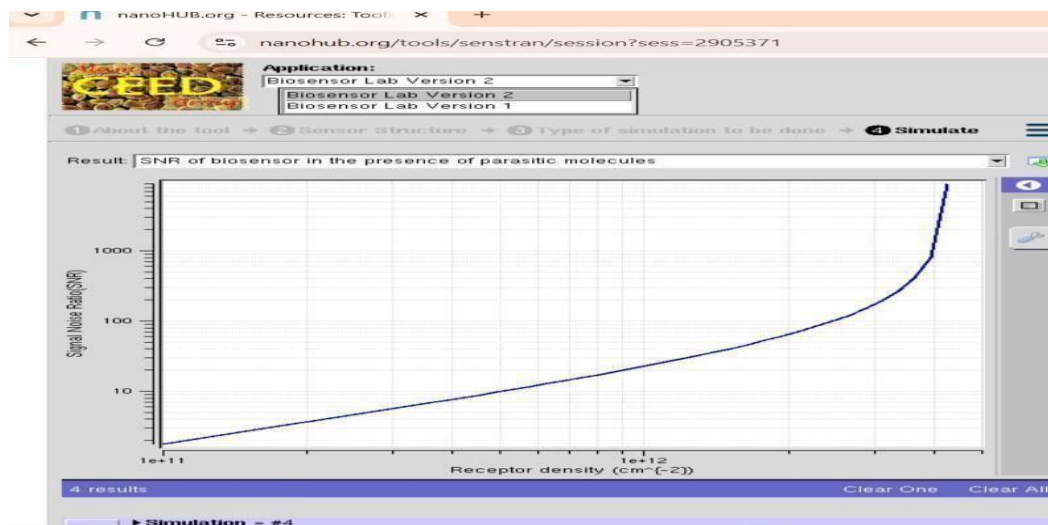
1. To simulate the threshold voltage of a BioFET-based biosensor as a function of electrolyte pH (4 to 8).
2. To analyse the transfer characteristics (I_{ds} vs. V_{fg}) of the BioFET for varying pH levels and interpret sensor performance.

Case Study 1: Comparative Analysis of Hybridized and Unhybridized BioFET Transfer Characteristics for Enhanced Biosensing Performance



Outcome: The graph compares the transfer characteristics (Drain Current, I_{ds} vs Gate Voltage, V_{gs}) of hybridized and unhybridized BioFET devices. The hybridized BioFET exhibits a noticeable shift in the transfer curve compared to the unhybridized device due to the binding of target biomolecules on the sensing surface. This shift indicates a change in the surface charge, which alters the channel conductivity and drain current. The results demonstrate that hybridization enhances the electrical response of the BioFET, enabling sensitive detection of biological molecules. This characteristic is useful for DNA, protein, and biomolecule sensing applications.

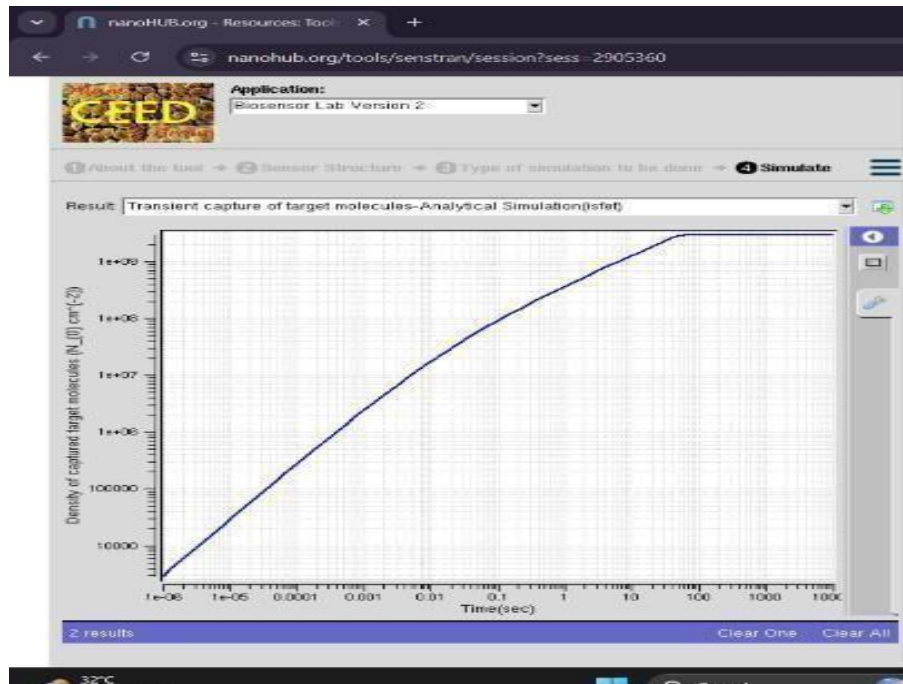
Case Study 2: Evaluation of Signal-to-Noise Ratio (SNR) Variation with Receptor Density in the Presence of Parasitic Molecules



Outcome:

The graph illustrates the variation of the Signal-to-Noise Ratio (SNR) with receptor density in the presence of parasitic molecules. As the receptor density increases, the SNR initially improves because more target molecules can bind to the sensor surface. Beyond a certain receptor density, the improvement becomes smaller due to the influence of parasitic molecules and saturation effects. The results indicate that selecting an optimal receptor density is important for achieving high detection sensitivity and reliable biosensor performance.

Case Study 3: Transient Analysis of Target Molecule Capture Dynamics in a BioFET-Based Biosensor



Outcome:

The graph shows the **transient response** of the BioFET during the capture of target molecules over time. Initially, the sensor response increases rapidly as target molecules bind to the receptor sites. As time progresses, the response gradually reaches a steady-state value, indicating that most of the available binding sites are occupied. This transient behavior provides information about the capture kinetics, response time, and sensing efficiency of the BioFET. Such analysis is important for evaluating the real-time performance of biosensors in biomedical and diagnostic applications.

Comparison and Overall Conclusion

The three BioFET characteristics were successfully analyzed using the NanoHub simulation tool. The transfer characteristics clearly distinguish the hybridized and unhybridized BioFET, demonstrating improved sensing performance after biomolecule binding. The SNR analysis shows that increasing receptor density enhances sensor performance up to an optimum value, after which parasitic molecules reduce the improvement. The transient analysis illustrates the real-time capture of target molecules and the stabilization of the sensor response over time. Among the three analyses, the hybridized BioFET transfer characteristic provides the clearest indication of biosensing performance, while the transient analysis gives valuable information about response speed and the SNR analysis helps optimize sensor design. Overall, the simulation demonstrates the effectiveness of BioFETs for highly sensitive, label-free, and real-time biosensing applications in medical diagnostics, environmental monitoring, and healthcare.